



Patients with advanced lateral osteoarthritis can return to sports and work after distraction arthroplasty plus lateral meniscal allograft transplantation combined with cartilage repair

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Abstract

Purpose This study aimed to report return to sports (RTS) and return to work (RTW) outcomes after distraction arthroplasty (DA) plus lateral meniscal allograft transplantation (MAT) combined with cartilage repair in active patients with advanced osteoarthritis. It was hypothesised that DA combined with lateral MAT would improve clinical and radiological outcomes and enable RTS and RTW for most patients.

Methods In total, 21 patients with advanced osteoarthritis (moderate to severe joint space narrowing at lateral edge on Rosenberg view and large cartilage defect of lateral femoral condyle) who underwent concomitant DA and MAT were retrospectively reviewed. Clinical outcomes were assessed using subjective knee scores [Lysholm score, International Knee Documentation Committee (IKDC) subjective knee score, and Tegner activity scale (TAS)] and functional tests (isokinetic extensor muscle strength test, single-leg vertical jump test, and single-leg hop for distance test). The rates of RTS, RTW, and satisfaction were evaluated. Radiological outcomes were assessed using magnetic resonance imaging (MRI) and X-ray (Rosenberg view).

Results The mean age at surgery and mean follow-up duration were 37.2 ± 5.9 years and 37.1 ± 5.4 months, respectively. All improvements in subjective scores were statistically significant ($p < 0.001$). The Lysholm score improved from 58.3 ± 8.1 to 84.3 ± 8.2 , the IKDC subjective score from 53.9 ± 10.4 to 78.0 ± 7.7 , and the TAS from 4.0 ± 0.5 to 5.1 ± 0.8 . The limb symmetry index (LSI) (%) of the extensor peak torque at an angular velocity of $60^\circ/\text{s}$ improved from 67.3 ± 19.2 to $88.4 \pm 20.3\%$ ($p = 0.001$). The LSI of the single-leg vertical jump test and single-leg hop for distance test improved from 62.8 ± 21.3 to $87.7 \pm 19.5\%$ and from 63.9 ± 20.8 to $85.5 \pm 18.1\%$, respectively (all, $p < 0.001$). All patients were able to return to any sports activity and work. However, 67% and 90.5% returned to their defined or desired level of sports activity and occupation intensity, respectively. Further, 76.2% were very satisfied or somewhat satisfied with the outcome at the last follow-up. The JSW increased by 0.8 ± 0.4 to 2.3 ± 0.6 mm ($p = 0.005$). In more than 90% of patients, > 50% of cartilage lesion was covered. The mean graft extrusion was 2.6 ± 1.0 mm.

Conclusion All patients who underwent distraction arthroplasty plus lateral MAT combined with cartilage repair returned to any sports and work at the last follow-up. Significant improvements in clinical outcomes and the radiographic joint space width were observed. However, the activity ability was somewhat reduced compared with the best preoperative level. This one-stage joint salvage treatment is a promising option for young and active patients with advanced OA who wish to return to high levels of sports activity and occupation intensity ($\geq$ Tegner activity scale 4).

Level of evidence III.

Keywords Knee osteoarthritis · Joint distraction · Meniscal allograft transplantation · Cartilage repair · Return to sports · Return to work

Introduction

Meniscal allograft transplantation (MAT) is indicated for treating meniscus-deficient knees in relatively young and active patients [33]. Detecting cartilage degeneration early

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and performing MAT before the joint space width (JSW) decreases can reduce graft extrusion and osteoarthritis (OA) progression [19, 28]. However, in practice, many patients considered for MAT already have advanced cartilage loss and joint space narrowing before the appearance of symptoms [19, 37].

To date, few studies have reported MAT results in young and active patients with advanced cartilage loss and joint space narrowing. Some studies have reported good results of simultaneously performing cartilage repair in patients with extensive cartilage defects by broadening the indication of MAT [30, 37]. However, the solution to joint space narrowing has been scarcely studied. Young patients with joint space narrowing inevitably have reduced knee function and often remain in the ‘treatment gap’ without proper treatment [8, 17].

In recent years, distraction arthroplasty (DA) has emerged as a viable treatment option for advanced OA in young patients. DA can delay the need for total knee arthroplasty because joint distraction can unload the articular cartilage by eliminating the contact pressure between the femoral condyle and tibial plateau [6, 38]. A satisfactory result can be expected when performing cartilage repair in the intra-articular environment biomechanically improved by DA [7, 12, 14]. The situation without meniscus should also be considered when implementing DA plus cartilage repair. When a meniscus is damaged or a significant portion is removed, the proprioceptive mechanism of the knee joint is impaired, and the function decreases [3, 39]. If MAT is performed simultaneously with DA and cartilage repair, functional benefits are expected as the proprioceptive mechanism and kinematics of the knee joint can be preserved [39].

Previous studies have shown that realignment osteotomy plus MAT with/without cartilage repair in unilateral compartment OA provides clinically considerable clinical and radiological improvements [26, 32]. In patients who are not indicative of realignment procedure, MAT and cartilage repair require other unloading procedures other than osteotomy, but there are few research reports on this. Robust evidence is needed on relevant outcomes after an unloading procedure (DA) plus MAT combined with cartilage repair to justify clinical benefits for patients with unilateral compartment OA without malalignment. As the objective of any surgical treatment is to provide functional recovery and practical benefits to patients, it is important to analyse the return to sports (RTS) and return to work (RTW) outcomes as measures of clinical efficacy in these patients [26].

Therefore, this study aimed to report the outcomes of DA plus lateral MAT combined with cartilage repair in active patients with advanced OA. Further, we aimed to evaluate the clinical efficacy of DA and its ability to allow RTS and RTW for patients. If RTS and RTW are possible after this

one-stage joint salvage treatment, this could further justify the unloading procedure plus MAT in these patients. It was hypothesised that DA combined with lateral MAT would improve clinical and radiological outcomes and enable RTS and RTW for most patients.

Materials and methods

Patient selection

In total, 28 patients who underwent DA plus lateral MAT combined with cartilage repair between 2015 and 2018 were enrolled. The surgical inclusion criteria were: (1) age ≤ 50 years, (2) lateral meniscus-deficient knee, (3) symptoms lasting ≤ 3 years, (4) a preserved central JSW ≥ 2 mm, (5) moderate to severe joint space narrowing at the lateral edge on Rosenberg view (JSW at lateral edge < 2 mm), (6) International Cartilage Repair Society (ICRS) grade 3 or 4, (7) mechanical axis of the hip–knee–ankle angle $< 3^\circ$, (8) no concomitant ligament injuries, and (9) consent to all examinations in the study. In particular, regardless of whether the national health insurance was applied or not, patients who fully agreed to this operation and who wanted activities of Tegner activity scale (TAS) 4 or higher related to sports and work after surgery were included. Based on the inclusion criteria, one patient with valgus deformity $> 3^\circ$, three patients who did not undergo follow-up magnetic resonance imaging (MRI), two patients with incomplete clinical data, and one patient with follow-up < 2 years were excluded. Eventually, 21 patients were included. This retrospective observational study was performed after obtaining approval from the appropriate ethics committee (KUMC 2021-07-007) and written informed consent from all patients.

Surgical technique

Lateral MAT with the keyhole technique and using fresh–frozen and non-irradiated grafts was performed by a single experienced surgeon (J.G.K.). In the keyhole technique, the carved bone bridge was inserted into the keyhole-shaped slot connecting the original attachments of the anterior and posterior horns of the lateral meniscus [22]. The size of the meniscal allograft was determined using a modified Pollard method [10, 31]. After arthroscopic preparation, a 4-cm longitudinal arthrotomy incision was made. The lateral femoral condyle (LFC) cartilage lesion was exposed, and the damaged cartilage and calcified cartilage layer were removed entirely. The area of cartilage lesion on LFC was routinely measured before stem cell implantation. Multiple holes (2–4 mm in diameter and 5 mm in depth) were created 2 mm apart using a CARTISTEM® drill bit (MEDIPOST

Co., Seongnam, Korea). In the case of a kissing lesion $> 1.5 \text{ cm}^2$, microfracture was performed using the conventional method. Then, a guide pin was inserted in the anteroposterior direction under the lateral eminence of the tibial articular surface and approximately 15° above the tibial slope [25]. After guide-pin placement, a 10-mm reamer (Arthrex, Naples, FL) was inserted to make the tibial hole. A keyhole-shaped slot was created using a customised osteotome and dilator (CelluMed, Seoul, Korea) via a previous tibial hole. After tibial preparation, the bone bridge was advanced into the keyhole-shaped slot, and the meniscal allograft was manually reduced via the arthrotomy incision under traction of the leading suture. The posterior one-third was repaired using the Fast-Fix 360 system (Smith & Nephew Endoscopy, Andover, MA, USA), and the middle horn was repaired with two superior and one inferior obliquely arranged sutures in an inside-out manner with double-armed needles. A mixture of sodium hyaluronate and human umbilical cord blood-mesenchymal stem cells (hUCB-MSCs) (CARTISTEM, MEDIPOST Co.) was implanted into the holes of the LFC from the base to the surface (Fig. 1). The anterior one-third was repaired with three or four direct sutures, using polydioxanone (PDS II; Ethicon, Somerville, NJ, USA) through the anterior arthrotomy incision in an outside-in manner. After the intra-articular procedure, knee joint distraction was performed using an Ilizarov external fixator (RISC RTO, Kurgan, Russia). The guide pin was inserted from the lateral

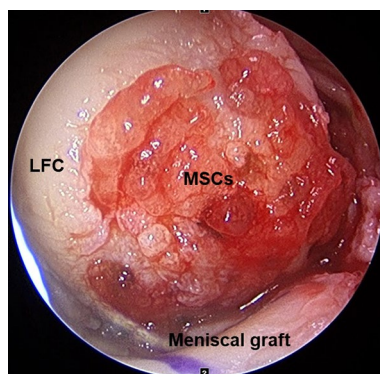


Fig. 1 A mixture of sodium hyaluronate and human umbilical cord blood-mesenchymal stem cells (MSCs) are implanted into the holes of the lateral femoral condyle (LFC) before the repair of the anterior one-third of the lateral meniscal allograft

epicondyle to the medial epicondyle parallel to the joint line, and half rings were fixed to the femur and tibia with two bone pins each bridging the knee joint. After the guide pin was removed, the knee joint distraction was gradually increased to 3–5 mm, and the range of motion was assessed. The half rings and pins were surgically removed 8 weeks after the knee joint distraction.

Postoperative rehabilitation

Active range of motion exercise, isometric quadriceps muscle strengthening exercise, and straight leg-raising exercise were started 3 days after lateral MAT. Patients were allowed to fully load the distracted knee, supported by crutches, for 12 weeks after surgery, initiating isokinetic exercise. Light running and squatting were allowed at 6 months, and RTS was allowed at 9 months. However, strenuous contact sports were prohibited until 10–12 months after surgery, according to muscle strength and knee function recovery.

Clinical evaluation

Clinical and functional tests were conducted before lateral MAT, 6 and 12 months after surgery, and every year after that. Data at 6 months preoperatively, 6 and 12 months postoperatively, and at the last follow-up were used to evaluate clinical outcomes. The Lysholm score, International Knee Documentation Committee (IKDC) subjective knee score, and Tegner activity scale (TAS) score were evaluated as subjective measures. Questionnaires about sports and work were administered to obtain information about RTS (including activity level) and RTW (including occupation intensity) [26, 27, 29]. Sports activity level was classified using the TAS based on the patients' self-reported activity level at four time points: preoperatively (within 5 years before DA plus MAT [the best activity level during the period was determined] and 6 months preoperatively) and postoperatively (1 year postoperatively and the last follow-up). Sports activity level was stratified according to TAS into low, medium, and high (Table 1). Patients were asked for each time point's highest sports activity level and sports frequency (times/week) and duration (hours/week). Patients were asked to qualify RTS at follow-up, compared to the best sports ability within 5 years before DA plus MAT. There were three

Table 1 Categorization of sport activity level ($N=21$)

Activity level	Sports
Low	Tegner activity scale 1–3: jogging, fitness sports, aerobics, yoga, pilates, golf, etc.
Medium	Tegner activity scale 4–6: cycling, skiing, badminton, baseball, spinning, cross country skiing, etc.
High	Tegner activity scale 7–10: soccer, basketball, tennis, baseball, squash, taekwondo, downhill skiing, etc.

categories: (1) return to participation (the patients may be participating in rehabilitation, training, or any sport, but at a level lower than their RTS goal), (2) return to sport (the patients have returned to their defined sports, but are not performing at their desired performance level, and (3) return to performance (the patients have gradually returned to their defined sports and are performing at or above their preinjury level) [2]. The rate of RTS was investigated at two-time points (1 year postoperatively and the last follow-up). As previously reported, occupation intensity was classified as heavy, moderate, light, and sedentary (Table 2) [26]. The patients reported their occupation intensity at four time points: preoperatively (during 5 years before DA plus MAT [the highest intensity during the period was determined] and 6 months preoperatively) and postoperatively (6 months postoperatively and the last follow-up). The rate of RTW was investigated at two time points (6 months postoperatively and the last follow-up). For each time point, the highest occupation intensity was asked. Moreover, work frequency (times/week) and duration (hours/week) were asked. Patients were asked the quality of RTW at follow-up, compared to the best work ability during the 5 years before DA plus MAT. There were three categories: (1) return to participation (The patients may be participating in any work but at a level lower than their RTW goal), (2) return to work (The patients have returned to their defined work but are not performing at their desired performance level, and (3) return to performance (The patients have gradually returned to their defined work and are performing at or above their preinjury level) [2]. Responses about patients' overall satisfaction with the surgery outcomes were assessed using an ordinal satisfaction scale (very satisfied, somewhat satisfied, neither satisfied nor dissatisfied, somewhat dissatisfied, and very dissatisfied) [29].

The isokinetic extensor muscle strength test was conducted at an angular velocity of 60°/s using a Biodex System IV dynamometer (Biodex Medical Systems, Shirley, NY, USA). The measurements were repeated three times at an angular velocity of 60°/s, and the average value was

calculated. For functional performance evaluation, the patients performed the single-leg vertical jump test and the single-leg hop for distance test [23]. Each test was repeated three times, and the best measurements were recorded. The limb symmetry index (LSI, %) was calculated by dividing the data of the affected limb by the data of the unaffected limb and multiplying the result by 100.

Radiographic evaluation

Radiological examination was performed before surgery, 6 and 12 months after surgery, and every year after that. Radiographs were obtained preoperatively and 24 months after lateral MAT to assess radiological outcomes. The central JSW was measured from the centre of the LFC to the centre of the lateral tibial plateau (LTP) on Rosenberg view [5]. The JSW at the lateral edge was measured from the edge of the LFC to that of the LTP on Rosenberg view. When a bony spur was present, it was excluded from the measurements (Fig. 2). Lower limb alignment was measured using the mechanical axis of the hip–knee–ankle angle in a standing anteroposterior scanography. The measurements of

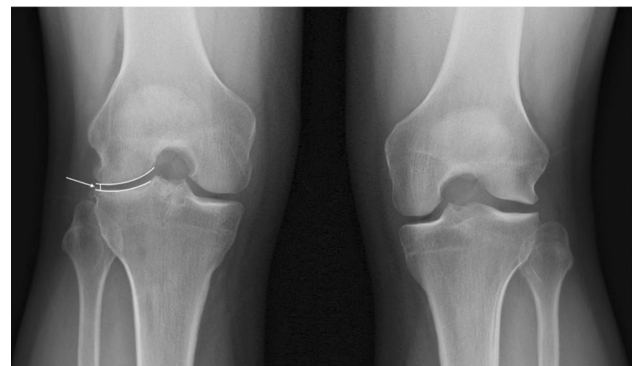


Fig. 2 The joint space width at the lateral edge (white arrow) is measured from the edge of the lateral femoral condyle to the edge of the lateral tibial plateau on Rosenberg view. When a bony spur is present, it is excluded from the measurements

Table 2 Categorization of occupation intensity ($N=21$)

Occupation intensity	Description
Sedentary	Tegner activity scale 1 (office work, hair dresser, etc.): sedentary work involves sitting most of the time but may involve walking or standing for brief periods
Light	Tegner activity scale 3 (shop assistant (light), preschool teacher, nursing, light domestic work, etc.): it requires (1) walking or standing to a significant degree; (2) constant pushing and/or pulling of controls; or (3) working at a production pace, in which an individual constantly pushes or pulls negligible weight
Moderate	Tegner activity scale 4 (shop assistant (heavy), truck driver, delivery worker, farmer, heavy domestic work, etc.): it involves exerting 20–50 lb of force occasionally, 10–25 lb of force frequently, or a negligible amount to 10 lb of force constantly
Heavy	More than Tegner activity scale 5 (building, forestry, fireman, police man, soldier, trainer, athlete, etc.): it involves exerting 50–100 lb of force occasionally, 25–50 lb of force frequently, or 10–20 lb of force constantly to move objects

JSWs were performed on a picture archiving and communication system (PACS) workstation (Centricity RA 1000; GE Healthcare, Chicago, IL, USA), and the agreement between orthopaedic surgeons (J.G.K. and D.W.L.) was assessed using data obtained 2 weeks apart. They were blinded to the patients' information.

The primary outcome was RTS evaluated by TAS, and the secondary outcome was the JSW at the lateral edge measured by X-ray.

MRI evaluation

All patients underwent MRI examinations using a 3.0-T system (Signa HD; GE Healthcare, Milwaukee, WI, USA). The articular cartilage status was assessed preoperatively and at 12 months postoperatively. The preoperative status of the cartilage was evaluated according to the ICRS grading system (grade 1, cartilage surface fibrillation; grade 2, < 50% loss of cartilage thickness; grade 3, > 50% loss of cartilage thickness without violation of the subchondral bone plate; and grade 4, full-thickness cartilage loss with exposed subchondral bone plate) [15]. Postoperative cartilage regeneration in the lateral compartment was grossly evaluated using the ICRS Cartilage Repair Assessment, as follows: grade I, complete or nearly complete coverage of the cartilage lesion (normal); grade II, 50–75% coverage (nearly normal); grade III, < 50% coverage (abnormal), and grade IV, no coverage (severely abnormal) [35, 36]. If an agreement was reached by at least three of four orthopaedic surgeons, the three matching results were selected as the final ICRS grade. For cases of an agreement by < 3 investigators, the final ICRS grade was determined through an open discussion among the four investigators. In addition, the modified Magnetic Resonance Observation of Cartilage Repair Tissue (MOCART) score was measured. To evaluate the lateral meniscal allograft extrusion, follow-up MRI examinations were performed at 12 weeks and 12 months postoperatively in patients who signed an informed consent form. Coronal graft extrusion was measured as the distance from the tibial plateau's superolateral aspect to the peripheral margin of the meniscal graft at the level of 0° on the sagittal plane using the crosslink tool of the PACS workstation (Centricity RA 1000, GE Healthcare). Pathological extrusion was defined as graft extrusion ≥ 3 mm [21].

Statistical analysis

Statistical analysis was performed using SPSS software (IBM SPSS Statistics 21; IBM Corp, Somers, NY, USA). Preoperative and postoperative parametric variables or non-parametric variables were compared using the paired *t*-test or Wilcoxon's signed-rank test. Statistical significance

was set at $p < 0.05$. Intra- and inter-examiner agreements were assessed using the kappa value and classified as excellent ($\kappa = 0.81$ – 1.00), substantial ($\kappa = 0.61$ – 0.80), moderate ($\kappa = 0.41$ – 0.60), fair ($\kappa = 0.21$ – 0.40), and slight (0 – 0.21) [18].

Results

The demographic data and detailed results are summarised in Table 3. The overall preoperative and postoperative data are summarised in the appendix (Table 7). The mean age at the time of surgery and mean follow-up duration were 37.2 ± 5.9 years and 37.1 ± 5.4 months, respectively. At the last follow-up, the Lysholm score had improved from 58.3 ± 8.1 to 84.3 ± 8.2 , IKDC subjective score from 53.9 ± 10.4 to 78.0 ± 7.7 , and TAS from 4.0 ± 0.5 to 5.1 ± 0.8 . All improvements in subjective scores were statistically significant ($p < 0.001$) (Table 4).

Return to sports and work

The rate of return to any sports activity was 90.5% (19 patients) at 1 year postoperatively and 100% at the last follow-up. However, the rate of return to their defined or desired level was 43% at 1 year postoperatively and 67% at the last follow-up. The rate of return to any work was 95.2% (19 patients) at 6 months postoperatively and 100% at the last follow-up. Yet, the rate of return to their defined or desired level was 76% at 1 year postoperatively and 90.5% at the last follow-up. Table 7 shows the trend of the TAS at four time points: preoperatively (within 5 years before DA

Table 3 Demographic and clinical characteristics ($N=21$)

	$N=21$
Age (years)	37.2 ± 5.9
Sex, male/female	16/5
Body mass index (kg/m^2)	23.6 ± 2.7
Diagnosis of discoid meniscus, n (%)	17 (81%)
Clinical follow-up duration (months)	37.1 ± 5.4
Lysholm score before DA plus MAT	58.3 ± 8.1
IKDC subjective score before DA plus MAT	53.9 ± 10.4
Best TAS during 5 years before DA plus MAT	6.2 ± 1.1
TAS 6 months before DA plus MAT	4.0 ± 0.5
ICRS grade in the LFC (1/2/3/4), n	(0/0/0/21)
ICRS grade in the LTP (1/2/3/4), n	(0/5/10/6)

DA distraction arthroplasty; MAT meniscal allograft transplantation; IKDC International Knee Documentation Committee; TAS Tegner activity scale; ICRS International Cartilage Repair Society; LFC lateral femoral condyle; LTP lateral tibial plateau

Table 4 Results of subjective knee scores, functional tests, and radiological findings at 6 months preoperatively and at the last follow-up ($N=21$)

	6 months preoperatively	Last follow-up	<i>p</i> value
Lysholm score	58.3 ± 8.1	84.3 ± 8.2	<0.001
IKDC subjective score	53.9 ± 10.4	78.0 ± 7.7	<0.001
Tegner activity scale	4.0 ± 0.5	5.9 ± 0.6	<0.001
LSI for isokinetic extensor strength at 60° (%)	67.3 ± 19.2	88.4 ± 20.3	0.001
LSI for single-leg vertical jump test (%)	62.8 ± 21.3	87.7 ± 19.5	<0.001
LSI for single-leg distance hop for distance test (%)	63.9 ± 20.8	85.5 ± 18.1	<0.001
Hip–knee–ankle axis angle ($^\circ$)	Valgus 0.1 ± 2.2	Varus 0.3 ± 2.1	0.007

IKDC International Knee Documentation Committee; LSI limb symmetry index; K–L Kellgren–Lawrence

plus MAT and 6 months preoperatively) and postoperatively (1 year postoperatively and the last follow-up). The participation rate in high activity level at the final follow-up did not reach the best level of the 5 years before surgery. However, the level of sports participation at three time points (6 months preoperatively, 1 year postoperatively, and the last follow-up) shifted from low to medium to high activity level. Likewise, the occupation intensity at three time points (6 months preoperatively, 1 year postoperatively, and the last follow-up) shifted over time from sedentary and light to moderate and heavy levels. Yet, it did not reach the best level of the 5 years before surgery (Table 5). Sports frequency and sport participation hours were lower at 1 year pre- and postoperatively, increasing at the last follow-up. Likewise, work frequency and work participation hours had increased again at the last follow-up. However, they did not reach the preoperative best level (Table 6).

Of the 21 patients, 76.2% were very satisfied or somewhat satisfied with the outcome at the last follow-up.

Functional and radiological outcomes

The extensor peak torque at an angular velocity of $60^\circ/\text{s}$ (Nm/kg) and the LSI (%) improved from 107.5 ± 31.1 to

176 ± 37.9 Nm/kg and from 67.3 ± 19.2 to $88.4 \pm 20.3\%$, respectively. The LSI (%) of single-leg vertical jump test and single-leg hop for distance test improved from 62.8 ± 21.3 to $87.7 \pm 19.5\%$ and from 63.9 ± 20.8 to $85.5 \pm 18.1\%$, respectively (Table 4).

Data on the central JSW and the JSW at the lateral edge on Rosenberg view are summarised in Fig. 3. The central JSW and the JSW at the lateral edge at 6 months postoperatively and at the last follow-up showed a significant increase compared with the preoperative data. The JSW at the lateral edge measured on Rosenberg view increased by 0.8 ± 0.4 to 2.3 ± 0.6 mm at a minimum follow-up of 2 years ($p=0.005$). Preoperative hip–knee–ankle angle was valgus $0.1 \pm 2.2^\circ$ and varus $0.3 \pm 2.1^\circ$ at last follow-up. There were 11 patients with valgus axis before surgery.

The mean graft extrusion at 1 year postoperatively was 2.6 ± 1.0 mm, and only 43% of patients had graft extrusion ≥ 3 mm. During operation, all patients showed grade 4 lesions in the weight-bearing portion of the LFC, and the mean chondral defect of the LFC was 5.1 ± 0.6 cm². Regarding cartilage regeneration, MRI at 12 months postoperatively demonstrated ICRS Cartilage Repair Assessment grade I in 7 (33.3%) patients, grade II in 12 (57.1%), and grade III in 2 (9.5%) (Fig. 4). There was no ICRS

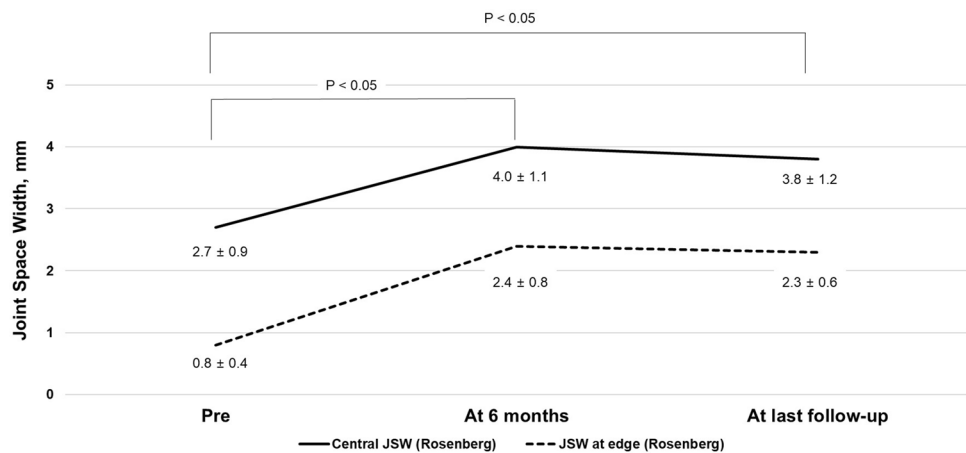
Table 5 Preoperative and postoperative sport and work activities ($N=21$)

	Best activity during 5 years before DA plus MAT	6 months before DA plus MAT	12 months after DA plus MAT	At last follow-up
Tegner activity scale	6.2 ± 1.1	4.0 ± 0.5	5.1 ± 0.8	5.3 ± 0.9
Sports activity level, <i>n</i> (%)				
Low	4 (19%)	11 (52.4%)	9 (42.9%)	6 (28.6%)
Medium	12 (57.1%)	10 (47.6%)	11 (52.4%)	13 (61.9%)
High	5 (23.8%)	0	1 (4.7%)	2 (9.5%)
Occupation intensity, <i>n</i> (%)				
Sedentary	0	3 (14.3%)	3 (14.3%)	1 (4.7%)
Light	2 (9.5%)	9 (42.9%)	7 (33.3%)	3 (14.3%)
Moderate	11 (52.4%)	7 (33.3%)	8 (38.1%)	11 (52.4%)
Heavy	8 (38.1%)	2 (9.5%)	3 (14.3%)	6 (28.6%)

DA distraction arthroplasty; MAT meniscal allograft transplantation

Table 6 Frequency and participation hours of sports and work ($N=21$)

	Best activity during 5 years before DA plus MAT	6 months before DA plus MAT	12 months after DA plus MAT	At last follow- up
Sports frequency (times/week), n				
No participation	0	2	2	0
1	3	10	9	7
≥ 2	15	8	8	12
≥ 4	3	1	2	2
Sport participation hours (hours/episode), n				
No participation	0	2	2	0
< 1	5	13	8	7
≥ 1	12	6	9	12
≥ 2	4	1	2	2
Work frequency (times/week), n				
No participation	0	0	1	0
1	0	0	0	0
≥ 2	6	11	10	9
≥ 4	15	10	10	12
Work participation hours (hours/day), n				
No participation	0	0	1	0
< 5	1	5	0	1
≥ 5	15	16	19	17
≥ 10	5	0	1	3

Fig. 3 After distraction arthroplasty plus lateral meniscal allograft transplantation, the central joint space width (JSW) and the JSW at the lateral edge on Rosenberg view showed significant improvements at 6 months and the last follow-up compared with the preoperative data

Cartilage Repair Assessment grade IV (Supplementary material). The mean modified MOCART score was 63.7 ± 14.6 .

The range of kappa values of intra- and inter-examiner agreement for all measurements was 0.81–0.92 (excellent). No limitation in the range of motion was observed during external fixator removal. However, two patients had a superficial infection and were adequately treated with oral antibiotics. No neurovascular injury associated with DA occurred.

Discussion

The important findings of this study are that all patients were able to return to any sports activity any work and that 67% and 90.5% were able to return to their defined or desired level of sports activity occupation intensity, respectively. Furthermore, as a one-stage joint salvage treatment, DA plus lateral MAT combined with cartilage repair significantly improved subjective clinical scores and radiographic JSW. Most patients reported being very

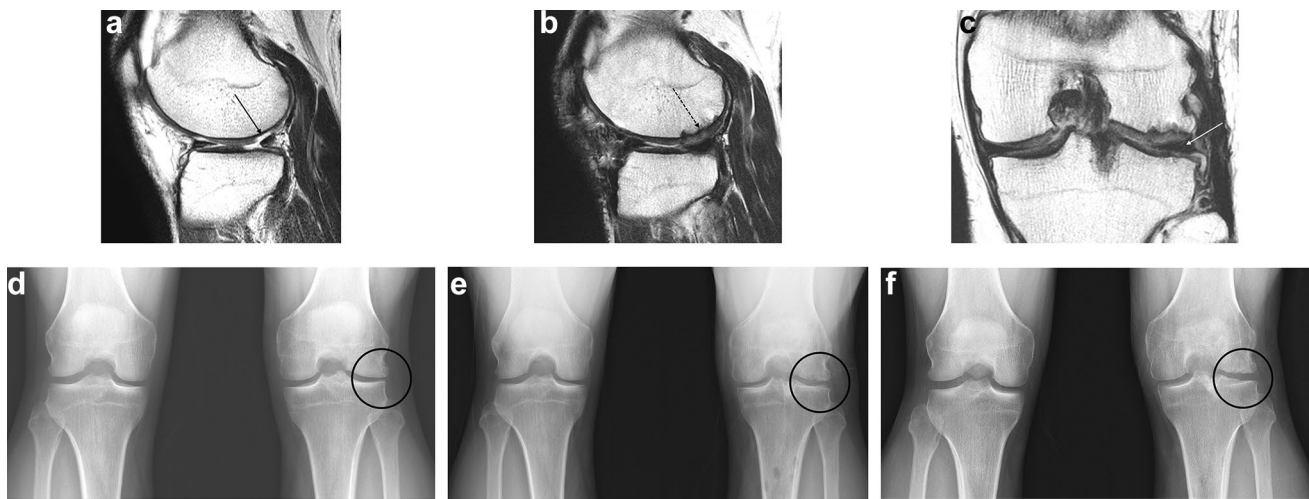


Fig. 4 A case of distraction arthroplasty plus lateral meniscal allograft transplantation with cartilage repair in the left knee. **a** The black arrow indicates a modified Outerbridge grade 4 cartilage defect in the lateral femoral condyle. **b** The dotted black arrow shows excellent regeneration of the cartilage in the lateral femoral condyle. **c** The white arrow indicates well maintained lateral meniscal allograft

without graft extrusion at 1 year of follow-up after distraction arthroplasty. **d** The joint space width (JSW) at the lateral edge was 0.4 mm. **e** The JSW at the lateral edge was 3.7 mm at 6 months postoperatively. **f** The JSW at the lateral edge was 4.9 mm at the last follow-up, which was higher than the measurement at 6 months postoperatively

satisfied or somewhat satisfied with the operation at the last follow-up. Based on these findings, it is believed that DA plus MAT combined with cartilage repair should be considered a new treatment concept of ‘biological reconstruction’, rather than a replacement for total knee arthroplasty. The early implementation of total knee arthroplasty will result in limitations on RTS and RTW, problems like wearing a prosthesis, and the need for early revision total knee arthroplasty.

As osteotomy is not indicated in all patients in this study, no treatment other than DA could be selected. A recent systematic review [11] proved that knee joint distraction could induce cartilaginous tissue regeneration and significant clinical improvement and that its benefit lasted up to ≥ 9 years. Jansen et al. [13] revealed that knee joint distraction could be a joint-preservation treatment in relatively young patients with advanced OA by comparing regular care and clinical trial. However, given the limited research on RTS and RTW after DA, a comparison with previous studies is difficult. Hoorntje et al. [8] analysed the results of DA and high tibial osteotomy from a randomised controlled trial. They reported that 79% and 94% of 16 patients achieved RTS and RTW status, respectively, after DA. However, there was a shift from participation in high and intermediate impact levels to participation in low impact levels, showing TAS decreased

from 5.0 to 3.5. All the patients demonstrated worse or much worse sports ability at the last follow-up than their best sports ability in their lifetime. Their study compared the clinical results of the two treatments in patients (mean age, 50.5 vs. 49.6 years) with advanced OA. It was not an appropriate DA case because they implemented DA in patients with an average malalignment of $> 6^\circ$ [8]. In this study, all patients could return to any sport and work as at the last follow-up, and 67% and 90.5% were able to return to their defined or desired level of sports activity and occupation intensity. Mean TAS decreased from 6.2 ± 1.1 to 5.3 ± 0.9 . Reporting overall RTS and RTW could be misleading because patients could not participate at their previous best level. In this regard, the outcomes of the current study seem to be superior to the previous research by Hoorntje et al. [8]. One reason was that this investigation was targeted at younger patients who did not have malalignment. It is presumed that MAT and cartilage repair were performed simultaneously and DA. MAT and cartilage repair could improve the proprioceptive mechanism of the knee joint. Thijs et al. [39] reported that a significant improvement of the proprioception after MAT was found at short-term follow-up (6 months), although there were no detectable knee functionality changes. They suggested that MAT could alter joint capsular tension and neural ingrowth into the transplanted

meniscal graft. Al-Dadah et al. [1] proved that patients with articular cartilage lesions (Outerbridge grade III or IV) had a significant proprioception deficit compared to normal controls. They postulated that cartilage restoration and joint space height could affect the proprioceptive mechanism. In this study, the functional proprioception test was not performed. Still, it is postulated that combined MAT and cartilage repair positively affected the knee function rather than the DA alone. Our study suggests that DA plus MAT combined with cartilage repair may offer an option for younger populations returning to high-level activities while delaying the timing of getting knee arthroplasty.

Our results are comparable to previously reported rates of RTS and RTW after high tibial osteotomy or distal femoral osteotomy in relatively young patients. Liu et al. [26] reported that 87.5% of 22 patients (aged 35.1 ± 8.1 years) could return to any sports activity by 9.7 months after high tibial osteotomy plus medial MAT. However, only 43.8% of the patients could return to the preinjury level of sports activities. In their study, all patients were able to return to any work. Contrastingly, only 88.9% returned to the preinjury level of occupation intensity. In their study, 62.5% of patients were satisfied with their sports activity level. The average frequency of participation in sports was decreased from 4.3 ± 1.9 to 3.0 ± 2.0 days. Hoorntje et al. [9] showed that eight out of ten patients achieved RTS and RTW status after distal femoral osteotomy. The mean age of 99 patients was 41.2 ± 8.1 years, and the median pre-symptomatic TAS was 4.0. The sports ability at the last follow-up was significantly reduced in 60% compared with their best lifetime ability. In their study, sports participation at four time points showed a shift over time from competitive or professional level to recreational level. Although sports frequency had increased at the final follow-up, it did not reach the pre-symptomatic level. The workload was decreased in 14%. Puzzitiello et al. [32] demonstrated that most of the 17 patients (mean age, 23.3 years) who underwent distal femoral osteotomy plus lateral MAT could return to sports at an average of 17 months. All patients could return to low activity level, and 73% and 53% to medium and high activity level, respectively. Furthermore, about 90% of patients were satisfied with their sport-related outcomes, although the frequency of participation was decreased. In this study, 67% and 90.5% could return to their defined or desired level of sports activity and occupation intensity, respectively. Regarding satisfaction with the overall condition at the last follow-up, 76.2% of 21 patients were very satisfied or somewhat satisfied with it. Thus, participation in sports and work

improved after the surgery, unlike the greatly decreased pre-operative level of activity and intensity, impacting wellness and overall satisfaction. These findings are consistent with those in high tibial osteotomy or distal femoral osteotomy. However, it cannot be concluded that any outcome is better because the criteria for RTS and RTW are different for each study. The criteria for classifying sports activity level and occupation intensity are also different. In this study, frequency and participation hours of sports and work also increased at the last follow-up, although they did not reach the best level during the 5 years preceding DA plus MAT. These findings are important for managing the patients' expectations preoperatively.

Structural and functional improvements were achieved through biological reconstruction with DA, lateral MAT, and cartilage repair. Although more evidence is needed to reveal the effects of DA on the outcomes of lateral MAT and cartilage repair, the 'unloading effect' of DA may be the most significant contributor to the structural improvement [42]. If the valgus malalignment is not severe ($< 3^\circ$), no surgical treatment other than arthroplasty is available for young and active patients because it is not an indication for joint unloading via osteotomy. In this situation, joint distraction can provide unloading of the affected compartment through a distraction device. Unloading of the affected compartment allows for restoring joint homeostasis, providing a healthier biomechanical and biological environment, and enabling tissue regeneration [4, 40–42]. In this study, the minimum JSW on Rosenberg view increased by 0.8 ± 0.4 to 2.3 ± 0.6 mm at a minimum follow-up of 2 years ($\Delta + 1.5$ mm). More than 90% of patients showed $> 50\%$ of cartilage regeneration after cartilage repair. This suggests a complementary relationship to the increased JSW of the lateral compartment. Further, the mean graft extrusion at 1 year was 2.6 ± 1.0 mm, and the major extrusion rate was only 43% in our study. In a recent meta-analysis, the pooled mean graft extrusion after lateral MAT was 3.01 mm, and the rate of major extrusion was 39% [20]. Although their meta-analysis results were mainly based on patients without OA progression and joint space narrowing, our findings on patients with advanced OA and joint space narrowing were considered comparable. We speculated that the unloading effect of distraction could reduce the occurrence of complications such as meniscal graft thinning, tear, and extrusion, which would have occurred if joint space narrowing had persisted.

Cartilage repair using stem cells is usually not implemented in extensive grade 4 lesions. However, cartilage repair can be considered in non-extensive grade 4 lesions

[16]. It is suggested that cartilage repair can be more aggressively considered in biomedical environments that have been improved through DA plus MAT. In this study, almost all patients showed satisfactory cartilage recovery after hUCB-MSC implantation during lateral MAT. hUCB-MSCs are known to have a high proliferation rate and karyotype stability after prolonged expansion. Therefore, they have greater chondrogenic potential than autogenous mesenchymal stem cells [34]. Song et al. [36] showed that hUCB-MSC implantation combined with high tibial osteotomy in patients aged > 60 years with a kissing lesion resulted in satisfactory regeneration of the cartilage. In their study, 42.9% of patients showed full cartilage recovery on second-look arthroscopy. Lee et al. [24] reported that hUCB-MSC procedure combined with high tibial osteotomy in patients with medial compartment OA with a kissing lesion showed ICRS Cartilage Repair Assessment grade I in 18.8% and grade 2 in 62.4%. Although unloading procedures such as osteotomy or joint distraction can result in fibrocartilage regeneration, the durability of the regenerated tissue is a major concern. Hence, hUCB-MSC implantation combined with joint distraction in patients with advanced OA (including a kissing lesion) may be considered. This structural recovery could directly translate to clinically meaningful improvements, especially in returning to sports activities.

This study is the first to report clinical and radiological outcomes at least 2 years after the unloading procedure (DA) plus MAT combined with cartilage repair. This study comprehensively analysed RTS and RTW and radiological outcomes of homogeneous populations undergoing DA plus MAT combined cartilage repair. Our results can potentially provide clinical evidence that DA plus MAT combined cartilage repair provides an acceptable level of RTS and RTW.

This study had several limitations: relatively small population size, possibly leading to a recall bias, and the absence of a control group. To diminish the recall, medical records were thoroughly reviewed. Moreover, as DA plus lateral MAT was developed to extend the indication of MAT, it is practically impossible to establish a control group without joint distraction when performing MAT in patients with

severe joint space narrowing. Further, the agreement on radiological measurements between two examiners was assessed using data obtained 2 weeks apart. This may result in memorial bias. In addition, the follow-up period was short. A minimum follow-up period of 2 years was insufficient to identify the progression of OA and graft extrusion after DA plus lateral MAT. Further, no validated questionnaire about RTS and RTW exists to ascertain them. The analysis of RTS and RTW seems to be somewhat subjective and overvalued. However, to evaluate the quality level of RTS and RTW, the degree of return was subdivided, and the frequency and participation times in sports and work were also investigated to obtain objective information. Finally, although it is not uncommon to perform cartilage repair at the time of lateral MAT, there is no clear indication for concomitant DA. More evidence is needed on the unloading of the lateral compartment with DA and addressing concomitant pathologies with lateral MAT and cartilage repair.

Conclusions

All patients who underwent DA plus lateral MAT combined with cartilage repair returned to any sports and work at the last follow-up. However, the activity ability was somewhat reduced compared with the best preoperative level. Moreover, significant improvements in clinical outcomes and in the radiographic joint space width were observed. This one-stage joint salvage treatment is a promising option for young and active patients with advanced OA who wish to return to high levels of sports activity and occupation intensity ($\geq$ Tegner activity scale 4).

Appendix

See Table 7.

Table 7 Preoperative and postoperative data

Case	Sex	Age at surgery	Best sports activity during 5 years before surgery	Sports activity at last f/u	Best work during 5 years before surgery	Work at last f/u	Defect size: MFC, cm ²	ICRS, CRA at LFC	Graft extrusion, mm	Pre-lower limb axis, (°)	Lower limb axis at last f/u, (°)	Pre-JSW at the lateral edge, mm	JSW at the lateral edge at last f/u, mm
1	M	36	Med. (badminton)	Med (badminton)	Building	Truck driver	5.1	II	2.8	-0.6	-0.4	0.5	2.8
2	F	23	Med (spinning)	Med (spinning)	Shop assistant (heavy)	Shop assistant (light)	4.7	II	0.5	1.4	1.6	0.7	2.1
3	M	30	High (soccer, competitive)	High (soccer, recreational)	Trainer	Trainer	4.8	III	3.5	-0.9	-1	1.1	3.4
4	M	38	Med (tennis)	Med (cycling)	Building	Office worker	6.1	I	2.9	1.8	2	0.7	2.7
5	M	42	High (soccer)	Mod (baseball)	Soldier	Soldier	5.6	II	2.4	2.2	2	0.8	2.3
6	F	46	Low (yoga)	Low (yoga)	Heavy domestic worker	Heavy domestic worker	5.9	II	3.8	-1.6	-2.1	0.6	2.4
7	F	44	Low (golf)	Low (golf)	Shop assistant (Heavy)	Shop assistant (Heavy)	4.3	II	2.1	0.3	0.6	0.4	2.1
8	M	30	Med (basketball)	Med (skiing)	Fireman	Fireman	4.6	II	3.7	-0.9	-1.4	1.1	2.9
9	F	42	Low (pilates)	Low (pilates)	Preschool teacher	Preschool teacher	5.8	III	4.5	-3.4	-4.2	0.9	1.9
10	M	41	Med (basketball)	Med (basketball)	Delivery worker	Delivery worker	4.5	I	0.7	0.9	0.7	0.7	2.2
11	M	34	Med (tennis)	Med (badminton)	Police man	Police man	6.2	I	2.1	2.1	1.8	0.3	3.1
12	M	35	Low (jogging)	Low (jogging)	Delivery worker	Delivery worker	4.5	II	2.6	-3	-3.2	1.7	1.8
13	F	29	Mod. (skiing)	Low (cycling)	Light domestic worker	Heavy domestic worker	5.3	II	3.6	2.3	2.1	0.7	1.5
14	M	41	High (basketball)	Mod (baseball)	Trainer	Trainer	5.8	I	2	-1.8	-1.9	0.6	2.8
15	M	43	Med (tennis)	Med (tennis)	Forestry	Delivery worker	4.8	II	2.6	2.9	3.1	0.8	2.1
16	M	35	High (soccer, competitive)	High (recreational)	Nurse	Nurse	4.1	I	0.7	-0.6	-1	0.5	3.5
17	M	38	High (taekwondo)	Med (spinning)	Athlete	Trainer	4.4	II	3.1	1.8	0.4	0.4	1.7
18	M	44	Med. (squash)	Low (jogging)	Farmer	Farmer	4.6	II	3	1.6	0.9	1.9	1.8
19	M	41	Med. (tennis)	Med (tennis)	Truck driver	Truck driver	5.2	I	3.2	3.7	1.9	1.2	2.1
20	M	37	Med (baseball)	Med (baseball)	Shop assistant (Heavy)	Shop assistant (Heavy)	5.8	II	3.1	-3.2	-3.8	0.8	1.5
21	M	33	Med (cycling)	Med (cycling)	Delivery worker	Delivery worker	5.1	I	2.1	-3.1	-3.4	0.7	2.4

f/u follow-up; MFC medial femoral condyle; ICRS CRA International Cartilage Repair Society Cartilage Repair Assessment; JSW joint space width; Med medium

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Declarations

Conflict of interest The authors have no conflict of interest to declare.

Ethical standards This retrospective observational study was performed after obtaining approval from the appropriate ethics committee (KUMC 2021-07-007).

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